

HYUNSOO KIM M.S.

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EDUCATION

Department of Artificial Intelligence, Korea University

Seoul, Republic of Korea

M.S. in Artificial Intelligence

Sep 2024 - present

- Advisor: Prof. Donghyun Kim (co-advised by Suhyun Kim)
- Research area: Diffusion Models, Data Synthesis, Concept Erasing

Department of Statistics, Inha University

Incheon, Republic of Korea

B.S. in Statistics and Software Engineering

Mar 2018 - Aug 2024

Two-year break for military service (Jul 2019 — Feb 2021)

PUBLICATIONS

1. Jungmyung Wi, **Hyunsoo Kim**, Donghyun Kim. HardSyn: Domain-Aligned Synthetic Data Generation for Hard-to-Classify Samples. *preprint*
2. Yongwoo Kim, **Hyunsoo Kim**, Sungmin Cha, Donghyun Kim. Safe and Consistent Image Generation with Synthetic Safe-Unsafe Paired Data Generation. *preprint*
3. **Hyunsoo Kim**, Wonjun Lee, Donghyun Kim[†], Suhyun Kim[†]. ReSafe: Enhancing Safety of Text-to-Image Diffusion via Post-Hoc Image Back Translation. *preprint*
4. Yujin Kim*, **Hyunsoo Kim***, Hyunwoo J. Kim. Suhyun Kim. When Model Knowledge meets Diffusion Model: Diffusion-assisted Data-free Image Synthesis with Alignment of Domain and Class. *International Conference on Machine Learning (ICML) 2025*.
5. **Hyunsoo Kim**, Donghyun Kim[†], Suhyun Kim[†]. Difference Inversion: Interpolate and Isolate the Difference with Token Consistency for Image Analogy Generation. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2025*.

EXPERIENCE

Korea Institute of Science and Technology (KIST)

Mar 2024 - Jun 2024

Undergraduate Researcher (*Advisor: Suhyun Kim*)

- Research on Model Inversion: Data-free image synthesis via optimizing text embedding on a text-to-image diffusion model.

Medical Image Innovation Lab, Seoul National University

Dec 2023 - Feb 2024

Undergraduate Intern (*Advisor: Wonjin Lee*)

- Research on conditional generation and image inpainting in medical domains using diffusion model.

Mathematical Intelligence Lab, Inha University

Aug 2022 - Sep 2022

Undergraduate Intern (*Advisor: Yoonsuk Hyun*)

- Studied Visual Language Models such as CLIP, CoCa, Flamingo.

Artificial Intelligence in Finance Lab, Inha University

Feb 2022 - Jul 2022

Undergraduate Intern (*Advisor: Doguk Kim*)

- Studied the basic concepts of Deep Learning, with a particular focus on the Transformer model.
- Participated in the M6 competition, a stock price prediction competition, and wrote baseline code using Transformer and GRU.

PROJECTS

Text-to-Emoji Generation with Diffusion Models

Naver Connects Foundation

Sep 2022 - Feb 2023